Quantifying Repair Risk in Crypto Option Surfaces

Tenghan Zhong
University of Southern California
Los Angeles, California, USA
tenghanz@usc.edu

Abstract

Option desks routinely adjust raw crypto option quotes into price surfaces that remain within bid–ask intervals and satisfy standard no-arbitrage restrictions across strikes before computing Greeks, hedge quantities, or risk measures. For the same quote panel, however, these constraints may admit multiple such surfaces, which we call *admissible repairs*, with different downstream outputs. We define the resulting variation as *repair risk* and propose *repair-risk validation*, a matched framework for quantifying dispersion in Greeks, hedge quantities, and risk measures.

The mechanism is finite-difference amplification: two admissible repairs within price distance ε can differ in local normalized-strike curvature by $4\varepsilon/h^2$, where h is local normalized-strike spacing. Motivated by this mechanism, we evaluate repair risk on Deribit BTC and ETH option chains, using L_1 and L_2 projections of midpoint prices onto the same admissible set as matched controls. Across 108 matched evaluations spanning 18 sampled calendar months, the median 90th percentile of the scaled spot-gamma gap for L_2 versus the $\rho = 10$ smoothness-penalized repair is nearly twenty times the matched L_1–L_2 control. We also instantiate NN-Proj, a neural-projection repair, to test whether learned repairs exhibit repair risk. Despite good held-out fit, its downstream outputs remain distinct from those of classical repairs. Across five seeds with the same training window and architecture, its curvature dispersion exceeds the matched L_1–L_2 control in 17 of 18 months ($p = 1.45 \times 10^{-4}$). Economically, for a USD 1 million risk reversal, switching from L_2 to the $\rho = 10$ repair shifts the median 90th-percentile hedge notional by 18.62 basis points of gross spot notional. VaR-style estimates also exhibit greater dispersion across repairs. The repair rule is therefore part of the risk model, not merely a feasibility check.

CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → **Model verification and validation**; *Machine learning*; • **Mathematics of computing** → Mathematical optimization.

Keywords

option surfaces, repair risk, no-arbitrage constraints, neural repair, model validation, crypto options, model risk, trustworthy AI



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1 Introduction

Raw crypto option quotes are rarely used directly in valuation and risk calculations. They can be wide, sparse, and violate no-arbitrage conditions across strikes. We therefore convert them to a normalized call-price vector and repair that vector to remain within bid–ask intervals and satisfy those conditions. We call any price vector satisfying these constraints an *admissible repair*, and write the feasible set as $\mathcal{A}$.

This step is usually treated as a price-feasibility screen. Most option-surface repair and arbitrage-free construction methods validate the repaired surface by price fit, feasibility, or smoothness. Recent neural surface methods are likewise often validated by fit and feasibility rather than downstream agreement. We instead ask a downstream question: after several classical or learned repairs satisfy the same feasibility constraints, do they agree on the Greeks, hedge quantities, and risk measures computed from the repaired surface?

We call this dispersion *repair risk*: for any scalar downstream quantity, it is the range of that quantity across admissible repairs of the same quote panel. This is an identification problem: a single feasible set can contain multiple admissible repairs that satisfy the same no-arbitrage constraints across strikes yet disagree on Greeks, hedges, or risk measures. Empirically, pairwise differences across the repair rules considered below provide observed lower bounds on the full feasible-set range. For local curvature, we compute the full range directly by linear programming.

A key mechanism behind this curvature range is finite-difference amplification. We show that two admissible repairs at sup-norm price distance ε can differ in local curvature with respect to normalized strike $x = K/F$ by $4\varepsilon/h^2$, where h is the local spacing in x. A repair negligible in price can therefore be material for curvature-based Greek estimates. Since no-arbitrage constraints across strikes do not uniquely select an admissible repair, any implemented repair requires an additional selection criterion.

Empirically, we study the consequences of this nonuniqueness on Deribit BTC and ETH option surfaces. The matched design holds the quote panel, feasible set, book templates, and downstream functionals fixed and varies only the repair rule. L_1 and L_2 projections serve as stable controls, while smoothness-penalized repairs separate further from L_2 as the penalty rises. NN-Proj, a learned repair obtained from a neural fit to total implied variance followed by projection into $\mathcal{A}$, remains distinct from both L_2 and the $\rho = 10$

smoothness-penalized repair, denoted Smooth_{10}, despite good out-of-sample fit in log total implied variance. The theory motivates the ranking score ($\text{RelSpread}/h^2$), which combines relative bid–ask spread with local normalized-strike spacing to rank slices by their potential for curvature amplification. The consequences are economic: the repair choice shifts hedge notionals on standardized books and produces material dispersion in VaR-style estimates.

This paper makes four contributions. First, we formulate *repair risk* as an identification problem: multiple admissible repairs of a single quote panel can satisfy the same constraints yet produce different Greeks, hedges, and risk measures. Second, a locally sharp construction gives the $4\varepsilon/h^2$ curvature separation, and linear programming characterizes the full feasible curvature range. Third, a separate 18-month replication using five training seeds shows that good predictive fit and projection into the same feasible set do not ensure stable outputs from a learned repair: neural refits still produce materially different admissible surfaces and curvature. Fourth, we quantify economic magnitude in BTC and ETH options by measuring how repair choice changes standardized-book hedges and VaR-style estimates.

Code Availability. The code repository is available at https://github.com/TenghanZhong/no-arb-code.

2 Related Work

Option-surface feasibility and repair. Finite-strike no-arbitrage, filtering, and repair methods characterize or project traded prices into feasible sets [8, 11, 21]. Arbitrage-free interpolation methods add construction choices beyond static feasibility [4, 17], while smoothing and parametric surface models impose additional regularity or low-dimensional structure [12, 13]. This literature treats feasibility, fit, and smoothness as central design goals. Our focus is repair risk after feasibility has already been imposed: among repairs within the same quote bounds and feasible set, how much can Greeks, hedges, and risk measures still differ?

Risk-neutral densities and smoothing sensitivity. The risk-neutral-density literature starts from the link between strike derivatives of option prices and state prices [6]. Nonparametric pricing and shape-restricted estimators show that density recovery depends on smoothing and no-arbitrage restrictions [2, 3]. Empirical implied-density work further shows that recovered distributions and moments can move with estimator choice and quote perturbations [5]. We recast this smoothing sensitivity as repair risk among admissible repairs within a fixed feasible set.

Crypto options and model uncertainty. Deribit inverse options and USD/coin accounting make repair and hedge measurement especially relevant in crypto option markets [19]. We view repair risk as a derivative-pricing form of model uncertainty [9]: the modeling choice is the rule that selects one admissible repaired surface from a fixed quote panel and feasible set.

Model multiplicity and learned surfaces. In machine-learning terms, the closest framing is underspecification and predictive multiplicity [10, 20]: multiple predictors can pass similar validation criteria yet differ on consequential outputs. We apply this lens to option-surface repair rules: feasibility is shared, while curvature, hedges, and risk measures may differ. Machine-learning methods have been used to learn option and implied-volatility surfaces [1, 7, 22]. Related work also includes physics-informed calibration of intraday implied-volatility surfaces under no-arbitrage constraints [15], generative and operator-based surface constructions [14, 24], and neural term structures as low-dimensional pricing representations [18]. Here, NN-Proj tests whether repair risk remains after projecting a learned target into the same feasible set.

Financial AI validation. The paper also relates to model-risk and trustworthy-AI frameworks in finance [23]. These frameworks emphasize that predictive fit alone is insufficient when modeling choices affect operational decisions. We extend this principle to learned option-surface repair: good fit and no-arbitrage feasibility need not ensure stable Greeks, hedge quantities, or risk measures.

3 Repair Risk and Feasible-Set Mechanism

This section isolates the local mechanism on a uniform normalized-strike grid; the empirical analysis uses the corresponding nonuniform three-point stencil. We proceed from feasibility to curvature amplification, its implications for Greek outputs, and bounds on local curvature repair risk.

3.1 Single-Maturity Feasible Set

Fix one maturity. Let N be the number of strike intervals, $h > 0$ the uniform normalized-strike grid spacing, x_0 the leftmost normalized strike, $x_j = x_0 + jh$, $j = 0, \dots, N$, and $c = (c_0, \dots, c_N) \in \mathbb{R}^{N+1}$ the normalized call prices. Define $(D_1 c)_j = (c_j - c_{j+1})/h$, $j = 0, \dots, N-1$, and $(D_2 c)_j = (c_{j-1} - 2c_j + c_{j+1})/h^2$, $j = 1, \dots, N-1$. Under our sign convention, the normalized price of the adjacent call spread is $c_j - c_{j+1} = h(D_1 c)_j$, so no arbitrage requires $0 \le (D_1 c)_j \le 1$. The corresponding butterfly-spread price is $c_{j-1} - 2c_j + c_{j+1} = h^2 (D_2 c)_j$; hence $(D_2 c)_j \ge 0$ enforces discrete convexity.

Let bid_j and ask_j denote the normalized bid and ask quotes, and let $b_j := \max\{\text{bid}_j, \max\{1 - x_j, 0\}\}$ and $a_j := \min\{\text{ask}_j, 1\}$ denote the effective price bounds. The single-maturity feasible set is

$$\mathcal{A} = \{c \in \mathbb{R}^{N+1} \ : \ b_j \le c_j \le a_j, \ 0 \le (D_1 c)_j \le 1, \ (D_2 c)_j \ge 0\}.$$

Each inequality is imposed over its natural index range. We apply $\mathcal{A}$ separately at each maturity and suppress the maturity index. Admissibility here isolates nonuniqueness among repairs satisfying the same no-arbitrage constraints across strikes; it does not impose cross-maturity no-arbitrage constraints. A vector is globally strictly feasible when $b_j < c_j < a_j$, $0 < (D_1 c)_j < 1$, and $(D_2 c)_j > 0$. Whenever nonempty, $\mathcal{A}$ is a compact convex polytope; the repair problems are convex programs, and each local D_2 range can be obtained from two linear programs.

3.2 Sharp Curvature Amplification

For $\varepsilon \ge 0$ and $c, \tilde{c} \in \mathbb{R}^{N+1}$ with $\|c - \tilde{c}\|_\infty \le \varepsilon$, the triangle inequality gives, for every interior i,

$$|(D_2 c)_i - (D_2 \tilde{c})_i| \le 4\varepsilon/h^2. \tag{1}$$

Theorem 1 shows that this bound is locally sharp within $\mathcal{A}$ itself: if c^0 is globally strictly feasible, the attaining repairs c^+ and c^- remain globally strictly feasible for all sufficiently small ε.

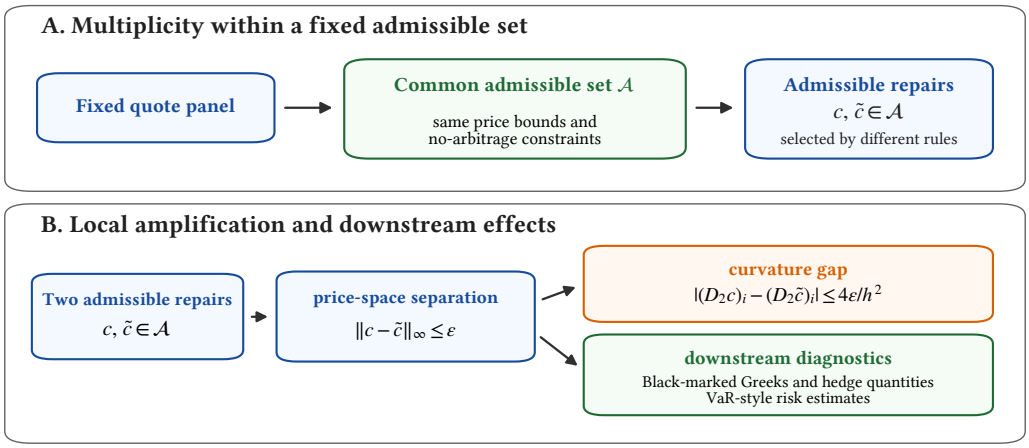


Figure 1: Fixed-set multiplicity and local amplification. A fixed quote panel can admit multiple repairs in the same admissible set; normalized-price distance ε can produce local curvature separation up to $4\varepsilon/h^2$. Greeks, hedge quantities, and VaR-style estimates are evaluated separately for each repair.

Theorem 1 (Sharp admissible curvature amplification). *Assume $N \ge 6$ and fix $i_\star \in \{3, \dots, N-3\}$. Let $I_0 = \{i_\star - 1, i_\star, i_\star + 1\}$, $I_1 = \{i_\star - 2, \dots, i_\star + 1\}$, and $I_2 = \{i_\star - 2, \dots, i_\star + 2\}$ be the affected price, first-difference, and second-difference stencils. These sets are bookkeeping: the three-point price perturbation changes three prices, four adjacent first differences, and five adjacent second differences. Suppose $c^0 \in \mathcal{A}$ has local slack $\kappa, s_1, s_2 > 0$:*

$$\begin{aligned} b_j + \kappa \le c^0_j \le a_j - \kappa \qquad & (j \in I_0), \\ s_1 \le (D_1 c^0)_j \le 1 - s_1 \qquad & (j \in I_1), \\ (D_2 c^0)_j \ge s_2 \qquad & (j \in I_2). \end{aligned}$$

Let $\eta := \min\{\kappa, hs_1/2, h^2 s_2/4\}$. For every $0 < \varepsilon \le 2\eta$, there exist $c^+, c^- \in \mathcal{A}$ with $\|c^+ - c^-\|_\infty = \varepsilon$ and $\left|(D_2 c^+)_{i_\star} - (D_2 c^-)_{i_\star}\right| = 4\varepsilon/h^2$. If c^0 is globally strictly feasible and $0 < \varepsilon < 2\eta$, then c^+ and c^- are globally strictly feasible.

Proof. Set $\alpha = \varepsilon/2$. Define $r_{i_\star - 1} = \alpha$, $r_{i_\star} = -\alpha$, $r_{i_\star + 1} = \alpha$, and $r_j = 0$ otherwise. Let $c^\pm = c^0 \pm r$. Then $\|c^+ - c^-\|_\infty = 2\|r\|_\infty = \varepsilon$.

On the affected rows, listed in increasing index order,

$$(D_1 r)_{I_1} = \frac{\alpha}{h}(-1, 2, -2, 1), \qquad (D_2 r)_{I_2} = \frac{\alpha}{h^2}(1, -3, 4, -3, 1).$$

Since $\varepsilon \le 2\eta$, $\|r\|_\infty = \varepsilon/2 \le \kappa$, $\|D_1 r\|_\infty = \varepsilon/h \le s_1$, and $\|D_2 r\|_\infty = 2\varepsilon/h^2 \le s_2$. Thus the affected price-bound, two-sided D_1, and D_2 constraints remain satisfied; all other constraints are unchanged, so $c^+, c^- \in \mathcal{A}$. Finally, $(D_2 r)_{i_\star} = 4\alpha/h^2$, so $(D_2 c^+)_{i_\star} - (D_2 c^-)_{i_\star} = 4\varepsilon/h^2$. If c^0 is globally strictly feasible and $\varepsilon < 2\eta$, all affected inequalities are strict and all other constraints retain their original strictness. □

Thus no-arbitrage feasibility alone does not eliminate the sharp local amplification (Figure 1).

3.3 Curvature in Risk Outputs

Let $c(\cdot)$ denote a twice differentiable normalized call-price curve whose values on the grid form the vector c. Under $V(S, K) = Sc(x)$, where V is USD option value, S is spot, K is strike, F is the forward, $q := F/S$, and $x := K/F = K/(qS)$, holding K, q, and $c(\cdot)$ fixed,

$$\Delta_S := \frac{\partial V}{\partial S} = c(x) - xc'(x), \qquad \Gamma_S := \frac{\partial^2 V}{\partial S^2} = \frac{x^2}{S} c''(x).$$

The curve-induced grid quantity $x_i^2 (D_2 c)_i / S$ inherits the bound in (1), scaled by x_i^2/S. The empirical analysis replaces D_2 by its nonuniform counterpart D_2^x. The empirical analysis evaluates reported Greeks with the Black formula under the inverse-option convention; these are separate functionals, and no algebraic equality with this curve-induced quantity is assumed.

3.4 Bounds on Local Curvature Repair Risk

Definition 1 (Local curvature repair risk). *For nonempty $\mathcal{A}$ and $i = 1, \dots, N-1$, define the local curvature repair risk by*

$$\mathcal{R}_{D_2}(i; \mathcal{A}) := \max_{c \in \mathcal{A}} (D_2 c)_i - \min_{c \in \mathcal{A}} (D_2 c)_i.$$

Because $\mathcal{A}$ is compact and the objective is linear, both extrema are attained and can be computed by linear programs.

Theorem 2 (Bounds on local curvature repair risk). *Assume $\mathcal{A} \ne \emptyset$. Fix $i_\star \in \{3, \dots, N-3\}$. If some $c^0 \in \mathcal{A}$ satisfies Theorem 1's local slack assumptions, then*

$$\mathcal{R}_{D_2}(i_\star; \mathcal{A}) \ge \frac{8\eta}{h^2} = \min\left\{\frac{8\kappa}{h^2}, \frac{4s_1}{h}, 2s_2\right\},$$
$$\eta := \min\{\kappa, hs_1/2, h^2 s_2/4\}.$$

Independently of this slack assumption, for any interior i and $\delta > 0$, if $a_j - b_j \le \delta$ for $j = i-1, i, i+1$, then $\mathcal{R}_{D_2}(i; \mathcal{A}) \le \min\{1/h, 4\delta/h^2\}$.

Proof. The lower bound follows from Theorem 1 with $\varepsilon = 2\eta$. For the upper bound, any $c, \tilde{c} \in \mathcal{A}$ satisfy $|c_j - \tilde{c}_j| \le \delta$ on the stencil, giving $4\delta/h^2$. Moreover, $(D_2 c)_i = ((D_1 c)_{i-1} - (D_1 c)_i)/h$, so the call-spread and convexity constraints imply $0 \le (D_2 c)_i \le 1/h$ for every $c \in \mathcal{A}$. Hence the feasible range is at most $1/h$. Combining these bounds proves the result. □

Remark 1 (Interpretation of the lower bound). For fixed κ, s_1, s_2, the lower bound $\min\{8\kappa/h^2, 4s_1/h, 2s_2\}$ is nonincreasing in h: denser grids raise the price-bound and call-spread terms until the convexity term binds.

4 Data and Validation Design

The theory highlights grid spacing h, box slack κ, the minimum slack s_1 to either call-spread bound, convexity slack s_2, and a local upper bound δ on effective price-interval widths. Rather than estimating the local slacks κ, s_1, s_2 separately, we use the linear-programming (LP) feasible range, which captures the joint effect of the effective price bounds and the imposed no-arbitrage constraints. The remainder of this section describes the data, feasible set, repair rules, and matched design used to evaluate repair risk empirically.

Data and normalization. The empirical study uses hourly Deribit BTC and ETH inverse (coin-settled) option chains represented in normalized call-equivalent prices; linear USDC-settled options are excluded. The main 12:00 UTC sample contains 375,894 quotes with timestamps, maturities, strikes, forwards, spots, prices, quality flags, and relative spreads. Under the inverse-option convention, the call premium c, quoted in units of the base currency, satisfies $c = \Phi_N(d_1) - (K/F)\Phi_N(d_2)$, with intrinsic value $\max\{1 - K/F, 0\}$. Here, K is strike, F is the recorded Deribit forward, Φ_N is the standard normal CDF, and d_1, d_2 are the usual Black quantities based on forward moneyness. Calls enter the repair grid directly. Let c_i and p_i denote normalized call and put premiums at strike K_i. Puts are mapped to the same normalized call-price coordinate by put–call parity, $c_i = p_i + 1 - K_i/F$, with bid and ask endpoints transformed consistently. The repair grid contains one call-equivalent observation per distinct strike within each maturity slice and is ordered by strike. After repair, put prices and put Greeks used in book templates are recovered by the same parity relation. The recorded forward F defines the common normalized-strike and price scale before repair and is held fixed across all matched repair comparisons.

Let $x_i := K_i/F$ be normalized strike. Empirically, we set $h_i = (x_{i+1} - x_{i-1})/2$ for interior observations, while endpoints use adjacent spacing. With $h_- = x_i - x_{i-1}$ and $h_+ = x_{i+1} - x_i$,

$$(D_2^x c)_i = \frac{2}{h_- + h_+}\left\{\frac{c_{i+1} - c_i}{h_+} - \frac{c_i - c_{i-1}}{h_-}\right\}.$$

The absolute row-sum norm is $4/(h_- h_+)$, reducing to $4/h^2$ on a uniform grid, so the corresponding nonuniform amplification scale replaces h^2 by $h_- h_+$. For interior observations, $h_i = (h_- + h_+)/2$, so $h_i^{-2} \le (h_- h_+)^{-1}$. We use $\mathrm{RelSpread}_i/h_i^2$, written $\mathrm{RelSpread}/h^2$, as a symmetrized proxy for the nonuniform amplification scale when ranking slices. Here $\mathrm{RelSpread}_i = (\mathrm{ask}_i - \mathrm{bid}_i)/\max\{|y_i|, 5\times 10^{-4}\}$, with y_i the normalized mid price.

We also restrict repair evaluations by maturity: $2 < \mathrm{DTE} \le 30$, where DTE denotes days to expiry. Marking and next-day book measures use $0 < \mathrm{DTE} \le 31$ with one extra marking day.

Evaluation units and slice selection. We use nested aggregation so inference is run at the calendar-month cluster level rather than the strike level. A slice is one option chain for a fixed currency, date, and maturity, observed at 12:00 UTC. Evaluations pool selected slices by currency, month, and selection group; currency-month cells aggregate selection groups within a currency and month; and calendar-month clusters aggregate BTC and ETH. For the VaR-style analysis, cells are additionally indexed by book template.

The classical design compares the smoothness path with the matched L_1–L_2 control in 108 evaluations: BTC and ETH over 18 sampled months, 2022-11 through 2026-05, each crossed with three slice-selection groups: high-score, random, and low-score. Only slices with a nonempty within-maturity admissible set ($\mathcal{A}$) enter the ranking. Slices are ranked by their 90th-percentile $\mathrm{RelSpread}/h^2$ score. The high-score and low-score selections take 50 slices from the top and bottom of this ranking, while the random selection draws 50 slices without conditioning on the score. The quote-spacing screen provides a simple threshold baseline for the ranking score. From the same evaluation pool, it selects wing quotes with short times to expiry when both relative spread and local spacing in normalized strike exceed their respective 75th percentiles within the relevant currency and DTE bucket. The $\mathrm{RelSpread}/h^2$ screen selects the same pool's top score decile. Every pair of repair rules is evaluated on the same selected slices, so selection changes where repair risk is measured while keeping the input data fixed across rules. For temporal coverage, the design begins with six anchor months (2022-11, 2023-03, 2023-06, 2024-03, 2025-01, and 2026-05). We then add 12 months sequentially, each chosen to maximize its minimum distance from the months already selected, with ties resolved in favor of the earlier month. Evaluation months and timestamps are fixed before repair outcomes are computed. The resulting design contains 36 currency-month cells and 18 calendar-month clusters. The targeted NN-Proj check asks whether its learned target yields a repair distinct from classical repairs. Its 10 matched-slice evaluations span 6 currency-month cells across 4 months: 2023-03 selection comparisons, 2023-06 high-score evaluations, and high-score out-of-sample windows.

Feasible set and repair rules. Feasibility is imposed separately at each maturity. Repaired prices must remain within the intersection of the quoted bid–ask intervals and the model-free bounds $[\max\{1-x_i, 0\}, 1]$, satisfy $0 \le (c_i - c_{i+1})/(x_{i+1} - x_i) \le 1$, and satisfy $(D_2^x c)_i \ge 0$ at every interior strike. In the empirical analysis, $\mathcal{A}$ denotes the within-maturity feasible set defined by these nonuniform-grid constraints. Let $y = (y_i)$ denote the vector of normalized midpoint prices, $\epsilon_p > 0$ the price-width floor, and $c^{\mathrm{NNProj,raw}}$ the unconstrained neural price vector. Set $W_1 = \mathrm{diag}(\max\{\mathrm{ask}_j - \mathrm{bid}_j, \epsilon_p\}^{-1})$. Let W_2 be diagonal, with squared weights $\max\{\mathrm{ask}_j - \mathrm{bid}_j, \epsilon_p\}^{-2}$ clipped within each slice at its 5th and 95th percentiles. The four repair rules below represent weighted projection, curvature regularization, and projection toward a learned target within the same feasible set. For $\rho \ge 0$, they are defined by

$$\begin{aligned}
\hat{c}^{L_1} &\in \arg\min_{c\in\mathcal{A}} \|W_1(c-y)\|_1,\\
\hat{c}^{L_2} &= \arg\min_{c\in\mathcal{A}} \|W_2(c-y)\|_2^2,\\
\hat{c}^{\rho} &= \arg\min_{c\in\mathcal{A}} \|W_2(c-y)\|_2^2 + \rho\|D_2^x c\|_2^2,\\
\hat{c}^{\mathrm{NNProj}} &= \arg\min_{c\in\mathcal{A}} \|W_2(c - c^{\mathrm{NNProj,raw}})\|_2^2.
\end{aligned}$$

We denote $\hat{c}^{\rho}$ by Smooth_ρ, with $\rho \in \{0.01, 0.1, 1, 10\}$. The ρ-grid defines the regularization path; ρ has no economic interpretation independent of the strike grid. Reported experiments use $\epsilon_p = 5\times 10^{-4}$ and the numerical floor $\epsilon_{\mathrm{num}} = 10^{-12}$. Since the L_1 minimizer

may not be unique, the deterministic HiGHS optimizer [16] under fixed solver settings and tolerances defines the implemented L_1 repair. Using linear interpolation over the overlapping normalized-strike range, a two-maturity calendar screen tests persistence under $c_{\text{long}}(x_g) \geq c_{\text{short}}(x_g)$.

Learned and parametric target repair checks. NN-Proj is a learned repair: a neural fit to total implied variance followed by projection. Its multilayer perceptron (MLP) fits $\log w$ on past samples, where σ_{imp} is implied volatility, $\mathcal{T}$ is year-fraction time to expiry, and $w := \sigma_{\text{imp}}^2 \mathcal{T}$ is total implied variance, using features $k = \log(K/F)$, $|k|$, k^2, k^3, $\log \mathcal{T}$, $\log(1 + \text{DTE})$, and $\log(1 + \text{RelSpread})$. The network standardizes features and uses hidden layers $(64, 64)$, tanh activations, Adam, an ℓ_2 weight penalty of 10^{-4}, at most 500 iterations, early stopping, and a 15% validation split. Predicted total implied variance is converted to $c^{\text{NNProj,raw}}$ and projected back onto $\mathcal{A}$. The targeted NN-Proj check uses up to 12 months of history; the seed replication tests sensitivity to training seeds after at least 12 observed training months. Both separate training and evaluation with a 7-day embargo. The reported coefficient of determination R^2 uses quote rows from the evaluation window.

Separate from the classical study, the seed replication uses slices with the highest RelSpread/h^2 scores in 18 months, each with a full 12-month lookback. Before examining seed outcomes, we excluded the 2025-01 and 2026-05 design pilots and selected the months by maximin spacing, retaining the first and last eligible months and resolving ties in favor of the earlier month. This design yields 36 currency-month cells and 180 refits. The D_2^x comparisons use slices on which all five projections succeed. Across refits, the training window, architecture, selected slices, feasible set, and projection operator are fixed; the replication measures randomized training variation. Finally, the stochastic-volatility-inspired (SVI) target check asks whether a standard parametric target also selects a distinct admissible repair. Its raw target $c^{\text{SVI,raw}}$ is obtained from a five-parameter raw SVI fit to midpoint total implied variance [13]. The SVI repair uses the smoothness objective above at $\rho = 0.01$, with y replaced by $c^{\text{SVI,raw}}$, leaving $\mathcal{A}$ unchanged.

Matched validation protocol. Figure 2 summarizes the matched protocol; the formal metrics are defined below.

Metrics. Strike. We compare repairs by the absolute spot-gamma difference scaled by the pairwise median absolute gamma within each slice. Let $\Gamma_i^{(m)}$ denote spot gamma at strike i under rule m. Operationally, each repaired normalized call price is inverted to implied volatility and evaluated with the same Black marking formula under the inverse-option convention to obtain USD spot delta and gamma. Let $\mathcal{S}(i)$ denote the evaluation slice containing strike observation i; all repair rules in a pair share this slice. For repair rules m, n and slice label $\lambda = \mathcal{S}(i)$, $\Delta^{\Gamma}_{mn,i} = |\Gamma_i^{(m)} - \Gamma_i^{(n)}|/M^{\Gamma}_{mn,\lambda}$, where $M^{\Gamma}_{mn,\lambda} = \max\{\text{median}_{j:\mathcal{S}(j)=\lambda,\ r\in\{m,n\}} |\Gamma_j^{(r)}|, \epsilon_{\text{num}}\}$. The median is over strikes j in slice λ and the two repair rules in the matched comparison. We report 90th percentiles (p90) within each evaluation.

Separately, the feasible-set curvature range uses the nonuniform operator D_2^x: $\mathcal{R}_{D_2^x}(i; \mathcal{A}) = \max_{c\in\mathcal{A}}(D_2^x c)_i - \min_{c\in\mathcal{A}}(D_2^x c)_i$, computed by two LPs for each selected interior strike. For the LP-range comparison, an observation is counted as an exceedance when its observed absolute D_2^x gap exceeds $\mathcal{R}_{D_2^x}(i; \mathcal{A})(1 + 10^{-5}) + 10^{-10}$. In Results, tables, and figures, empirical D_2^x quantities are abbreviated as D_2.

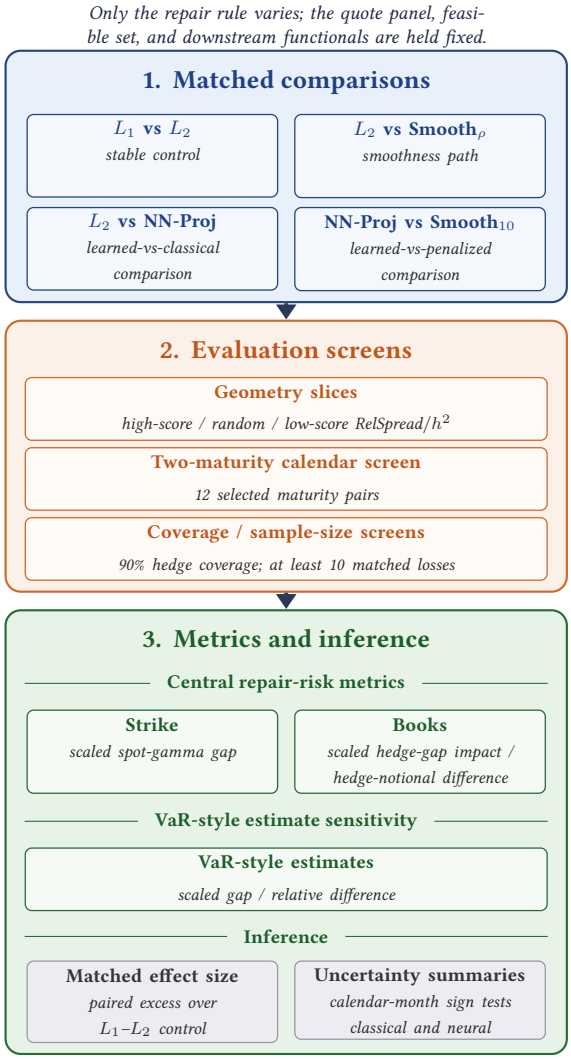


Figure 2: Matched validation protocol. The protocol fixes the quote panel, feasible set, and downstream functionals, then compares repair rules through evaluation screens, repair-risk metrics, and inference.

Books and VaR-style estimates. The treatment of missing next-day quotes can affect sample composition and measured book losses [25]. We therefore require 90% next-day coverage for evaluation-level hedge and book summaries. VaR-style cells require at least 10 matched next-day losses. Let C_k, P_k denote nearest available call/put legs at target $k = \log(K/F)$. Standardized templates use fixed gross spot notional: an ATM straddle $C_0 + P_0$, a put spread $P_{-0.05} - P_{-0.10}$, a risk reversal $C_{0.05} - P_{-0.05}$, and a put butterfly $P_{-0.10} - 2P_{-0.05} + P_0$. These templates target level, wing-slope, skew, and wing-convexity exposures, respectively.

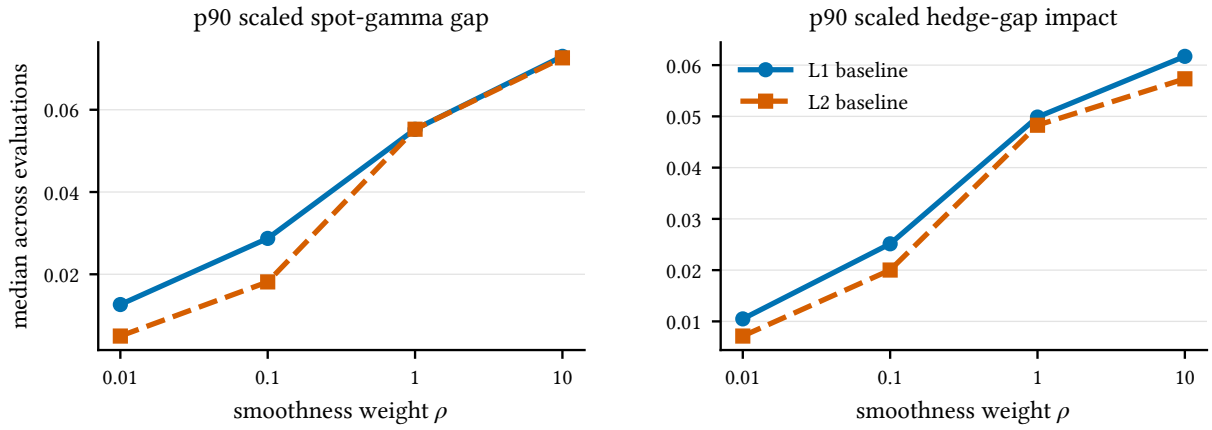


Figure 3: Classical smoothness path. Baseline-to-smoothness dispersion is nondecreasing in ρ under both the L_1 and L_2 baselines.

For a fixed book and evaluation cell at date t, let $B_t^{(m)}$ be the repaired book value and $H_t^{(m)}$ the hedge position, defined as minus the book delta computed from repaired USD spot deltas and fixed leg weights. Let S_t denote spot at date t. The primary hedge metric is scaled hedge-gap impact: the absolute hedge gap times the absolute next-day spot move, divided by the largest absolute value among the common and repaired book valuations for the two rules, with the denominator floored at USD 1. Separately, hedge notional gaps are $10^4|H_t^{(m)} - H_t^{(n)}|/\sum_\ell |v_\ell|$ bps and are summarized by their median p90; $S_t \sum_\ell |v_\ell|$ is gross spot notional and v_ℓ are fixed leg quantities.

Write $\Delta\Pi_{t+1}^{(m)}$ for next-day hedged book change, including book revaluation, hedge P&L, and transaction cost. The common-scenario calculation fixes the book legs, observed t and $t+1$ option marks, spot move, and denominator across repairs, so only the hedge varies. When the book is revalued using each repair, that repair determines the current and next-day book values, hedge, and denominator. The corresponding normalized loss is $-\Delta\Pi_{t+1}^{(m)}/\max\{|B_t^{(m)}|, 1\}$, with transaction cost set to zero in reported experiments. Together, the two calculations distinguish hedge changes from full revaluation.

VaR-style estimates are empirical 95% quantiles of next-day normalized losses matched across repairs within each book-template and selection-group cell. Because these cells contain 10–50 next-day observations, the estimates are descriptive measures of tail loss rather than regulatory backtests. Let u index a valid cell, identified by currency month, book template, and selection group, and let $v_{m,u}$ be its VaR-style estimate under repair rule m. The pairwise VaR-style gap is scaled as $|v_{m,u} - v_{n,u}|/\max\{|v_{m,u}|, |v_{n,u}|, \epsilon_{\text{num}}\}$. For comparisons with L_2, we also report the absolute relative difference

$$\omega_{m,u} := \frac{|v_{m,u} - v_{\text{L}_2,u}|}{\max\{|v_{\text{L}_2,u}|, \epsilon_{\text{num}}\}}.$$

Inference. For matched comparisons, we compute paired excesses over the matched L_1–L_2 control. Since selection groups within a currency month are correlated, we first aggregate paired excesses within currency-month cells. For classical smoothness, we then take the median across BTC and ETH cells within each calendar month before the sign test. For the neural seed replication, the primary excess is the median pairwise p90 D_2 gap on slices where all five projections succeed minus the matched L_1–L_2 control; BTC and ETH are aggregated by calendar-month median. We report two-sided exact sign tests at the calendar-month level and, for classical smoothness, the corresponding one-sided test.

5 Results

5.1 Smoothness Selects Distinct Admissible Repairs

Main result. Table 1 collects the headline measures. Since L_1 and L_2 use different weighting rules, their small spot-gamma gap provides the matched projection-rule control. Across the 108 classical control evaluations, the median p90 scaled spot-gamma gap is 0.0039 for L_1 versus L_2.

Along the implemented ρ-path, smoothness regularization changes only the selection criterion within the fixed feasible set (Figure 3). The median p90 scaled spot-gamma gap for L_1 versus Smooth_ρ rises from 0.0126 at $\rho = 0.01$ to 0.0730 at $\rho = 10$. For L_2 versus Smooth_ρ, it rises from 0.0050 to 0.0726. The scaled hedge-gap impact follows the same pattern: its median p90 rises from 0.0105 to 0.0617 for L_1 and from 0.0071 to 0.0573 for L_2.

The paths indicate a systematic smoothness effect: spot-gamma gaps are nondecreasing in 108/108 evaluations under both baselines; hedge-gap impacts are nondecreasing in 95/108 under L_1 and 96/108 under L_2. At $\rho = 10$, the evaluation-level excess of L_2–Smooth_{10} over the L_1–L_2 control is 0.0686. After currency-month and calendar-month aggregation, the excess remains positive in all 18 months, with median 0.0651 and exact sign test $p = 3.81 \times 10^{-6}$ one-sided (7.63×10^{-6} two-sided).

Robustness checks. The median calendar-month excess is similar across the six anchor months and 12 added months (0.0678 versus 0.0639). The two-maturity calendar screen on 12 selected maturity pairs leaves the observed dispersion essentially unchanged: the median post-screen/pre-screen dispersion ratio is 1.000. An 18-month SVI-target check extends the result to a standard parametric smile target: L_2 versus the SVI-target repair is 0.0834, and its excess over the matched L_1–L_2 control is positive in 108/108 evaluations. The SVI-target projection succeeds on 5,376/5,400 preselected problems.

Table 1: Summary of main empirical results.

Claim / comparison	Measure and sample	Estimate	Diagnostic
L_1 vs. L_2 control	Median evaluation-level p90 scaled spot-gamma gap; 108 evaluations	0.0039	Small control gap
L_2 vs. Smooth_{10}	Same measure; 108 evaluations and 18 calendar-month clusters	Gap 0.0726; excess 0.0686	Median month excess 0.0651; 18/18 positive; two-sided $p = 7.63 \times 10^{-6}$
LP feasible D_2 range	Observed D_2 gap versus LP range; 51,189 high-score interior strikes	Spearman 0.8722; log–log $R^2 = 0.6112$	LP-bound exceedances: 57/51,189 initially; 0 after tighter-tolerance recomputation
Targeted NN-Proj check	Spot-gamma gaps on successful projections; 10 evaluations, 500/500 problems	L_2–NN-Proj 0.1890; NN-Proj–Smooth_{10} 0.1957; control 0.0108	Positive in 6/6 currency-month cells and all 4 calendar months
Five-seed replication	Pairwise D_2 gaps; 36 high-score currency-month cells and 18 calendar months	1.3822 versus control 0.0719; paired excess 1.2003	17/18 months positive; two-sided $p = 1.45 \times 10^{-4}$; slices shared across seeds: median 100%, minimum 80.9%
Downstream books / VaR-style estimates	Hedge: 42 risk-reversal evaluations; VaR-style: 432 common-scenario and 417 repair-specific cells	Hedge 18.62 bps; VaR-style 0.0111/0.0933 and 0.0904/0.3882	Full-revaluation gaps are descriptively larger

Notes: Spot gamma entries are median p90s across evaluations; VaR entries are median/p90 over valid cells: common scenario first, full revaluation under each repair second.

5.2 Locating and Bounding Repair Risk

To locate candidate slices, we compare two screens across 108 evaluations and classical repair pairs: median p90 scaled spot-gamma gaps are 0.0457 on 103 nonempty quote-spacing screens and 0.0651 on 95 nonempty RelSpread/h^2 screens. Within each evaluation, regressing $\log(1+\Delta^\Gamma)$ on log relative spread and $\log(1/h_i^2)$ gives both coefficients positive in 100/108 evaluations but weak fit (median $R^2 = 0.0290$), so we use RelSpread/h^2 only as a slice-ranking score.

We then compute the feasible curvature range from $\mathcal{A}$: two LPs per selected interior strike give the D_2-range. In high-score L_2–Smooth_{10} slices across 36 currency-month cells and 18 calendar months, this range has Spearman correlation 0.8722 with observed D_2 curvature gaps and log–log $R^2 = 0.6112$. Under the stated numerical criterion, 57/51,189 observations exceed the computed LP range. Recomputing these 57 cases with tighter solver tolerances yields no exceedance at an absolute tolerance of 10^{-9}. Figure 4 supports the feasible-set mechanism: the score locates slices, while the LP range measures the curvature variation allowed by $\mathcal{A}$ and bounds realized gaps.

5.3 Learned Repairs and Sensitivity to Training Seeds

The NN-Proj results show that predictive fit and no-arbitrage feasibility are distinct requirements. Across 10 evaluations, training samples contain 10,719 to 45,412 option quotes, and held-out R^2 for log total implied variance has median 0.941, with range 0.858 to 0.973. Despite this predictive performance, before projection, NN-Proj prices fall outside the bid–ask intervals for a median 55.9% of quotes, with a median price mean absolute error (MAE) of 2.21 bid–ask widths. Prediction is evaluated in log total implied variance, whereas feasibility is imposed in normalized price units, so projection into $\mathcal{A}$ is required. Slice eligibility is fixed before repair, and all 500 preselected NN-Proj surfaces project successfully within solver tolerance; subsequent comparisons therefore concern admissible repairs, while next-day book coverage is assessed separately.

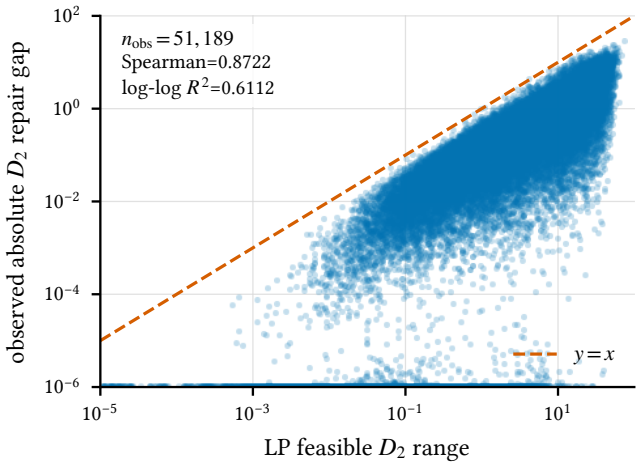


Figure 4: Feasible curvature range and observed repair gap. The plot compares the LP feasible D_2 range with the observed D_2 gap for L_2–Smooth_{10} high-score slices; the dashed line is the identity line, and values below 10^{-6} are shown at the plotting floor.

Learned repair versus classical repairs. On matched slices with successful projections, sharing the same feasible set does not make NN-Proj coincide with L_2 or Smooth_{10}. Across the 10 evaluations, the median 90th percentile of the scaled spot-gamma gap is 0.1890 for L_2 versus NN-Proj and 0.1957 for NN-Proj versus Smooth_{10}, compared with 0.0108 for the matched L_1–L_2 control. Currency-month paired excesses are positive in 6/6 cells for both comparisons and remain positive in 4/4 calendar months using the same median rule. Median p90 scaled hedge-gap impact is 0.1149 for L_2 versus NN-Proj and 0.1328 for NN-Proj versus Smooth_{10}. Figure 5 shows that both the spot-gamma gap and hedge-gap impact are larger for NN-Proj versus Smooth_{10} than for NN-Proj versus L_2.

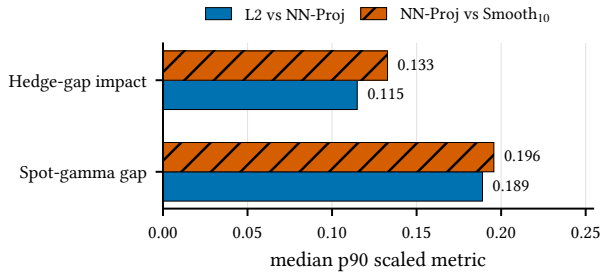


Figure 5: Targeted learned-repair check. Post-projection NN-Proj **is compared with** L_2 **and** Smooth_{10} **in spot-gamma and book hedge measures.**

Seed sensitivity. In the 18-month seed replication, the median p90 nonuniform D_2 gap across seed pairs on slices where all five projections succeed is 1.3822, versus 0.0719 for the matched L_1–L_2 control, with median paired excess 1.2003. The excess is positive in 34/36 currency-month cells and 17/18 months after BTC/ETH aggregation (two-sided exact $p = 1.45\times10^{-4}$). These common slices comprise a median 100% and a minimum 80.9% of selected slices across windows. Despite similar price fit after projection, curvature gaps remain large: the worst refit's root-mean-square error (RMSE), scaled by bid–ask width, exceeds the best by at most 8.17% across windows (median 1.10%).

5.4 Downstream Hedge and VaR-Style Effects

We first report standardized-book results across all classical repair comparisons, followed by separate results for L_2–Smooth_{10} and L_2–NN-Proj. Figure 6 reports common-scenario absolute relative differences in VaR-style estimates for Smooth_{10} relative to L_2. The largest hedge differences occur in risk reversals, whose call and put legs lie on opposite wings. Across 171 standardized book cells that pass the coverage screen, median next-day pair coverage is 0.9787 and the median p90 scaled hedge-gap impact is 0.0080. ATM straddles have a median p90 impact of 0.0002, while risk reversals reach 0.0330 because repairs affect their two legs differently.

Across the 42 risk-reversal evaluations passing the coverage screen, the median p90 absolute hedge-notional gap between L_2 and Smooth_{10} is 18.62 bps, equivalent to USD 1,862 of hedge position per USD 1 million gross spot notional. Across valid cells for the four standardized books, the scaled pairwise VaR-style gap has median/p90 0.0111/0.0933 under the common-scenario calculation (432 cells) and 0.0904/0.3882 when the book is revalued using each repair (417 cells).

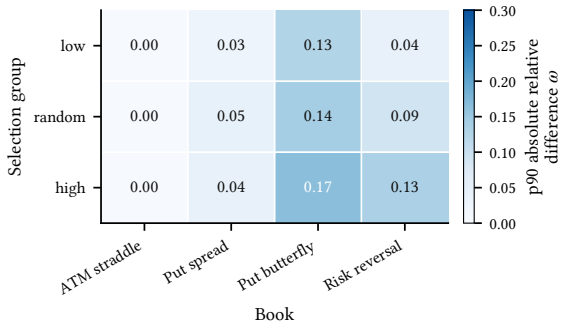


Figure 6: Absolute relative differences in common-scenario VaR-style estimates. Cells report the 90th percentile of the absolute relative difference $\omega_{m,u}$ **for** Smooth_{10} **relative to** L_2**.**

Under the absolute relative difference $\omega_{m,u}$, the put butterfly book shows the largest common-scenario VaR-style sensitivity, unlike the hedge-gap ranking above. For the matched L_2–NN-Proj comparison, the 90th percentile is 0.1226 and the median is 0.0597. The absolute denominator $|v_{\text{L}_2,u}|$ is floored at 10^{-12}; no reported common-scenario cell triggers this floor. Under repair-specific revaluation, the corresponding 90th-percentile differences are 0.7400 for L_2–Smooth_{10} and 0.6100 for L_2–NN-Proj among their respective valid cells. These values are larger than when only the hedge varies, but the sets of valid cells differ.

6 Conclusion and Implications

Option-surface price feasibility is an input to model validation, not a substitute for it. Remaining within bid–ask intervals and satisfying static no-arbitrage across strikes do not ensure stable downstream Greeks, hedge quantities, or risk measures. The matched design shows that smoothness penalties and projection toward a learned target change downstream risk quantities while holding the raw quote panel, feasible set, and downstream functionals fixed. Theorem 2 gives the upper bound $\min\{1/h, 4\delta/h^2\}$ on local curvature repair risk, where δ bounds the three effective interval widths and h is normalized-strike spacing. Together with the locally sharp construction in Theorem 1, this shows that narrow effective price intervals need not imply tight curvature identification on dense strike grids. The upper bound holds for any pair of admissible repairs, including smoothness-penalized repairs, learned repairs, and seed refits of the same learned model, while the quote panel and feasible set stay fixed. The classical comparisons cover 18 sampled months from November 2022 to May 2026, with additional evidence from neural seed refits and a two-maturity calendar screen.

The implication extends beyond the sampled BTC and ETH windows: when quote bounds and no-arbitrage constraints admit multiple repairs, the repair rule is part of the risk model. Validation should therefore assess dispersion in Greeks and hedge quantities across matched repairs, alongside price fit and no-arbitrage feasibility. The LP curvature range quantifies the local variation permitted by the full admissible set and helps identify slices that warrant closer review. Future work can examine how repair choice affects stochastic volatility calibration and subsequent pricing and hedging.

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